Fadhil Muhammed N C

AI / Machine Learning Engineer

Kerala, India | +91-8136931448 | ncfadhil@gmail.com

linkedin.com/in/fadhilmuhd | github.com/fadnc | fadhilmuhammednc.vercel.app

Professional Summary

AI/ML engineer (MSc Computer Science, Data Analytics, 2026) with hands-on experience in Generative AI, LLM agents, RAG, NLP, and tool calling, plus a recent agentic AI internship. Built production features for a multi-tenant voice agent, including a usage-metering billing engine and an MCP server exposing booking, knowledge-base, and CRM tools, and cut meeting-assistant response latency about 2.4x. Also built deep learning ICU risk-prediction models over 61,000+ stays and a LangGraph multi-agent system deployed as four services on AWS ECS. Available to start immediately.

Technical Skills

Languages: Python, SQL

LLMs and Agents: LLM agents, agent orchestration, LangGraph, MCP, tool calling, prompt engineering, QLoRA fine-tuning

Generative AI and NLP: Generative AI, RAG, FAISS, cross-encoder reranking, NLP, Hugging Face

Machine Learning: Deep Learning, Computer Vision, PyTorch, scikit-learn, XGBoost, LightGBM, SHAP, Pandas, NumPy, YOLOv8, OpenCV

Data and Math: Statistical Analysis, Probability, Linear Algebra, Feature Engineering, Predictive Modeling

Backend and Cloud: FastAPI, PostgreSQL, REST APIs, Docker, Git, AWS (ECS, ECR), GitHub Actions, MLflow

Experience

AI Intern - Agentic AI Developer Jul 2026 – Aug 2026

DifferentByte (ByteFlow), Kochi, India

- Built a wallet-based billing engine for a multi-tenant voice agent that meters per-session LLM, STT, TTS, and avatar usage and burns credits in real time, with booking, knowledge-base, and CRM tools exposed through a native MCP server.
- Implemented self-service multi-tenancy, a booking flow with CRM lead capture, and a reranked RAG knowledge base for a white-label chatbot live on a client website.
- Extended a meeting assistant across Google Meet, Zoom, and Teams with calendar-based scheduling for unattended joins.
- Migrated the assistant's LLM to Cerebras and tuned reasoning settings, cutting measured response latency about 2.4x without observed quality loss; began building an unattended, auto-scored HR interview mode.

ML Research Project — Smart ICU Assistant Jan 2026 – May 2026

Digital University Kerala | MIMIC-III | GitHub

- Built an ICU risk-prediction pipeline over 43.6 GB of MIMIC-III (61,532 stays), engineering 81 features per timestep and training deep learning (LSTM, Transformer) and gradient-boosted (XGBoost, LightGBM) models on chronological splits for 6 clinical tasks.
- Led sepsis, vasopressor, and ventilation prediction; a 4-model ensemble per task (AUROC-squared weighting, F1 threshold tuning) reached AUROCs of 0.77, 0.81, and 0.85, up to 0.03 above the best single model.
- Delivered a FastAPI dashboard with SHAP explanations and 8 rule-based alert types to triage patients by risk level.

Projects

Agentic Resume Intelligence System GitHub

- Built an agentic system that helps job seekers find gaps between their resume and a job description and generates targeted rewrites, using a 4-agent LangGraph pipeline (parser, JD analyst, gap analyst, rewriter).
- Conditional routing skips the rewriter on matches scoring below 30, saving 1 of 4 LLM calls and returning full analyses in 7–15 s.
- Implemented dual retrieval (RAG) over a per-request FAISS index and 4 prebuilt corpora of 29 curated hiring documents, grounding every agent in real hiring guidance instead of generic LLM output and removing startup indexing.
- Deployed 4 AWS ECS Fargate services through a GitHub Actions CI/CD pipeline, enabling push-to-deploy releases.

Finance Text-to-SQL Assistant GitHub

- Built an assistant that translates plain-English financial questions into executable SQL by adapting TinyLlama 1.1B to the domain with QLoRA (4-bit NF4, rank 16; 2.25M trainable parameters, 3 epochs, 4 GB VRAM) on 245 synthetic samples across 8 SQL pattern families over a six-table schema.
- Developed an execution-based evaluator to distinguish SQL syntax, schema hallucination, and logical errors; achieved 88.5% execution accuracy (23/26) on a small, prompt-grouped holdout, with all 3 failures attributed to logic errors.

Audio-Visual Emotion Recognition GitHub

- Developed a multimodal emotion recognition system to investigate whether combining vocal and facial cues improves emotion classification over a single modality, using PyTorch and attention-based fusion of audio spectrograms and facial features to classify 8 emotions.
- Achieved 73.3% accuracy, 0.723 macro-F1, and 0.958 macro OvR AUROC on the actor-held-out RAVDESS test set using face-focused audio-visual fusion.
- Built a FastAPI application for video-based emotion prediction, integrating MediaPipe face detection, facial geometry analysis, confidence scores, and GPU-accelerated inference.

Education

MSc Computer Science, Data Analytics, Digital University Kerala (SGPA: 8.2) 2024 – 2026

BSc Mathematics, Farook College (Autonomous), Kozhikode 2021 – 2024